



National University of Computer and Emerging Sciences, Lahore Campus

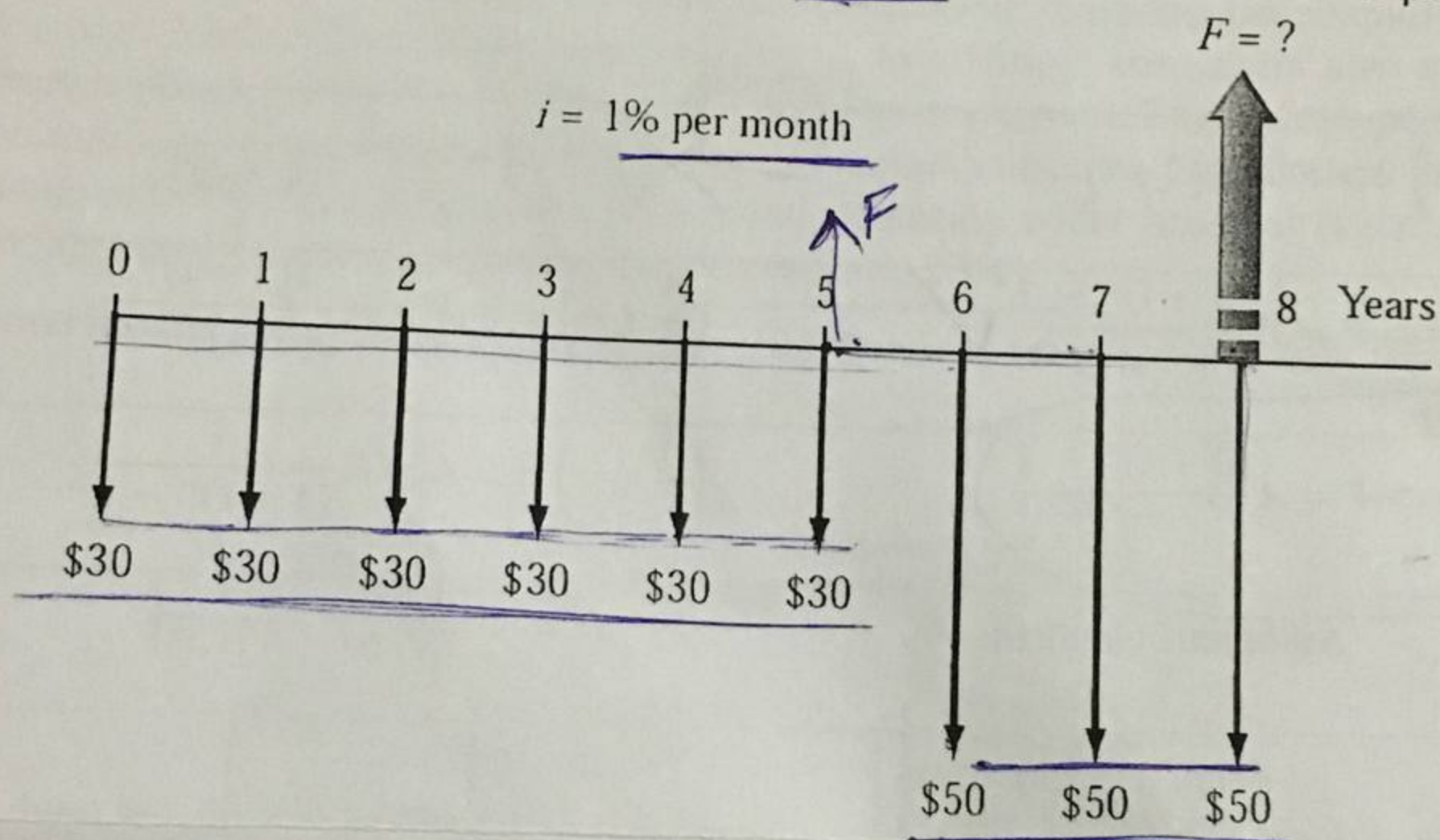
Course: Engineering Economics
Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 12-April-18
Section: ALL
Exam: Mid II

Course Code: MG217
Semester: Spring 2018
Total Marks: 30
Weight: 20%
Page(s): 2
Questions: 3

Instruction/Notes: Start each question from new page, show all steps, and assume if something is missing in paper.

Q1 For the cash flow diagram shown, solve for F , using an interest rate of 1% per month.

[10]
CLO
02



341.65

Q2 National Homebuilders, Inc., plans to purchase a new cut-and-finish equipment. Two manufacturers offered the estimates below.

[10]
CLO
03

	Vendor A	Vendor B
First cost, \$	-15,000	-18,000
Annual M&O cost, \$ per year	-3,500	-3,100
Salvage value, \$	1,000	2,000
Life, years	3	4

Determine which vendor should be selected on the basis of a present worth comparison, if the MARR is 15% per year.

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Course:	Engineering Economics	Course Code:	M247 M622
Program:	BS(Electrical Engineering)	Semester:	Spring 2018
Duration:	60 Minutes	Total Marks:	20
Paper Date:	26-Feb-18	Weight	20%
Section:	ALL	Page(s):	2
Exam:	Mid I	Questions	3

Instruction/Notes: Start each question from new page, show all steps, and assume if something is missing in paper.

- Q1 CJK Rheosystems makes high-performance rotational viscometers capable of steady Shear and yield stress testing in a rugged, compact footprint. How much could the company afford to spend now on new equipment in lieu of spending \$200,000 one year from now and \$300,000 three years from now, if the company uses an interest rate of 15% per year. (draw the cash flow diagram) [5]
CLO
01

371.16

- Q2 Morris Glass Company has decided to invest funds for the next 5 years so that development of "smart" glass is well funded in the future. This type of new technology glass uses electrochrome coating to allow rapid adjustment to sun and dark in building glass, as well as assisting with internal heating and cooling cost reduction. The financial plan is to invest first, allow appreciation to occur, and then use the available funds in future. All cash flow estimates are in \$1000 units, and the interest rate expectation is 8% per year. [10]
CLO
02

Year 1 through 5: Invest \$7000 in year 1, decreasing by \$1000 per year through year 5. $P_T = 20576$

Year 6 through 10: No new investment and no withdrawals. $P_0 = 27763$

Year 11 through 15: Withdraw \$20,000 in year 11, decreasing 20% per year through year 15.

Determine if the anticipated withdrawals will be covered by the investment and appreciation plans, also draw the cash flow diagram. 1197.31

- Q3 Find the numerical value of the following factor (A/P, 14%, 15). 0.1628

[5]